



Software Testing and Quality Assurance

(05302438)

Yadagiri Rama Deepak,
Assistant Professor
Computer Science & Applications



Module-1

1. Software Testing Fundamentals(W-15)

Introduction, Testing objectives, Test information flow, Testing lifecycle, Test Cases, Designing test cases.

Learning Outcome

1. Understand the Fundamentals of Software Testing

- Define software testing and explain its significance in the software development lifecycle.
- Identify the role of testing in ensuring software quality.

2. State and Explain Testing Objectives

- Describe the main objectives of testing, such as defect detection, quality assurance, and validation & verification.
- Differentiate between error, fault, failure, and defect.

3. Analyze Test Information Flow

- Explain how information such as requirements, test plans, and defect reports flow through the testing process.
- Understand the interaction between testing artifacts (test plans, test cases, test scripts, and test reports).

Learning Outcome

4. Describe the Software Testing Lifecycle (STLC)

- Outline the various phases of STLC: Requirement analysis, Test planning, Test case development, Environment setup, Test execution, Test cycle closure.
- Explain the entry and exit criteria for each phase.

5. Understand and Develop Test Cases

- Define what a test case is and explain its importance in structured testing.
- List the key components of a test case (Test case ID, Objective, Inputs, Expected results, etc.).

6. Design Effective Test Cases

- Apply techniques such as equivalence class partitioning, boundary value analysis, and decision table-based testing to design efficient test cases.
- Identify valid and invalid inputs to uncover defects.

7. Improve Critical Thinking and Analytical Skills

- Develop the ability to design test cases that improve code coverage and effectiveness.
- Assess the quality and completeness of test cases.

Introduction

- I. Introduction to Software Testing
- II. Importance of Software Testing
- III. Objectives of Testing
- IV. Principles of Software Testing
- V. Software Testing Life Cycle (STLC)
- VI. Types of Testing
- VII. Levels of Testing
- VIII. Manual vs Automation Testing
- IX. Common Testing Techniques
- X. Defect Life Cycle
- XI. Tools Used in Software Testing

Introduction

Definition:

Software Testing is the process of evaluating a software system or application to identify whether it meets the specified requirements and to detect any errors or bugs.



Introduction

Small Example : Industry Example: Banking Application

Scenario:

A bank launches an **online money transfer feature** in its mobile app.

Requirement:

When a user transfers ₹1000 from Account A to Account B, the amount should:

- Be **deducted** from Account A ✓
- Be **credited** to Account B ✓
- Generate a **transaction receipt** ✓

During Testing:

The tester finds that:

- ₹1000 is **deducted from Account A**,
- But Account B **does not receive** the amount ✗



Introduction

Why is Software Testing Important?

Ensures quality and reliability
Prevents failures and bugs in production
Reduces maintenance cost
Improves user satisfaction
Helps meet compliance and regulatory standards

Introduction

Objectives of Software Testing

To detect defects

To validate software meets requirements

To ensure reliability and performance

To gain confidence in the software product

Introduction

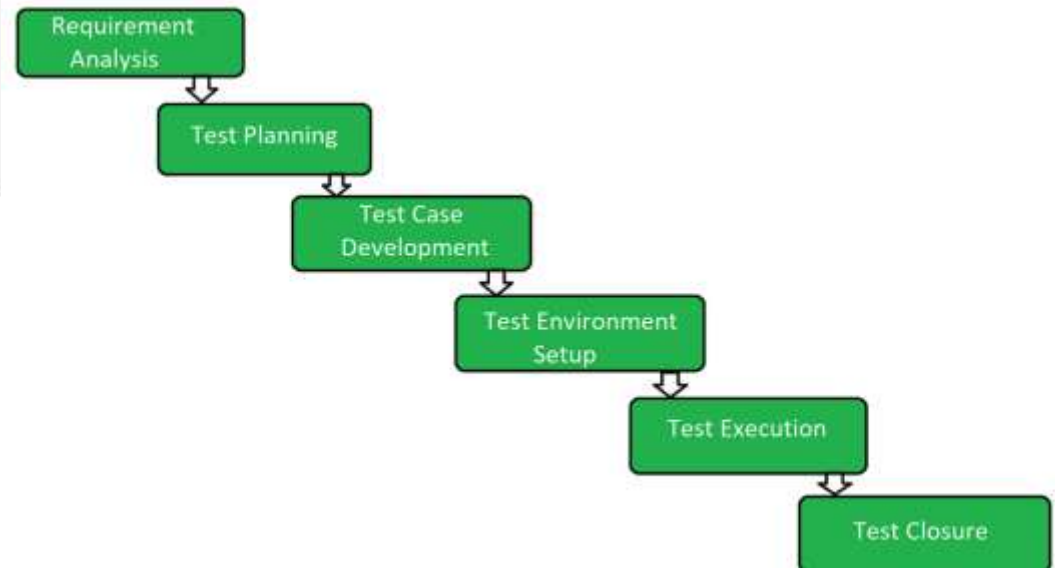
Principles of Software Testing

1. Testing shows presence of defects
2. Exhaustive testing is impossible
3. Early testing saves time and cost
4. Defect clustering
5. Pesticide paradox
6. Testing is context dependent
7. Absence-of-errors fallacy

Introduction

Software Testing Life Cycle (STLC)

Requirement Analysis
Test Planning
Test Case Development
Environment Setup
Test Execution
Test Closure



Introduction

Types of Testing

Functional Testing

Non-Functional Testing(e.g., Performance, Security, Usability)

Maintenance Testing(e.g., Regression, Retesting)

Introduction

Levels of Testing

Unit Testing – testing individual components

Integration Testing – testing interfaces between modules

System Testing – testing the entire application

Acceptance Testing – validating software with user requirements



Testing objectives

Test objectives are verbal statements detailing the goals of the tests in a test module. They are not mandatory, but in many project's they can help clarify why certain actions are taken and checks are performed.

Testing objectives

5 Typical Objectives of Testing





Testing objectives

Characteristics of effective study objectives

- Specific: Clearly define what you want to achieve.
- Measurable: Ensure progress can be tracked.
- Attainable: Set realistic goals.
- Relevant: Align objectives with your product or business goals.
- Time-Specific: Set a clear timeline for achieving the objectives



Testing objectives

Testing objectives can include various testing techniques like regression testing, root cause analysis, and verification of functional requirements. These objectives aim to confirm that the software meets the specified requirements, ensuring both verification and validation of the software product



Test information flow

What is information Flow

Information flow is the exchange of information among people, processes and systems within an organization. When you have employees working across different locations, devices and departments, it can be difficult to keep everyone on the same page.

Test information flow

The testing process is broken down into several crucial stages: requirements analysis, test planning, test design, test run, test analysis, reporting, and maintenance



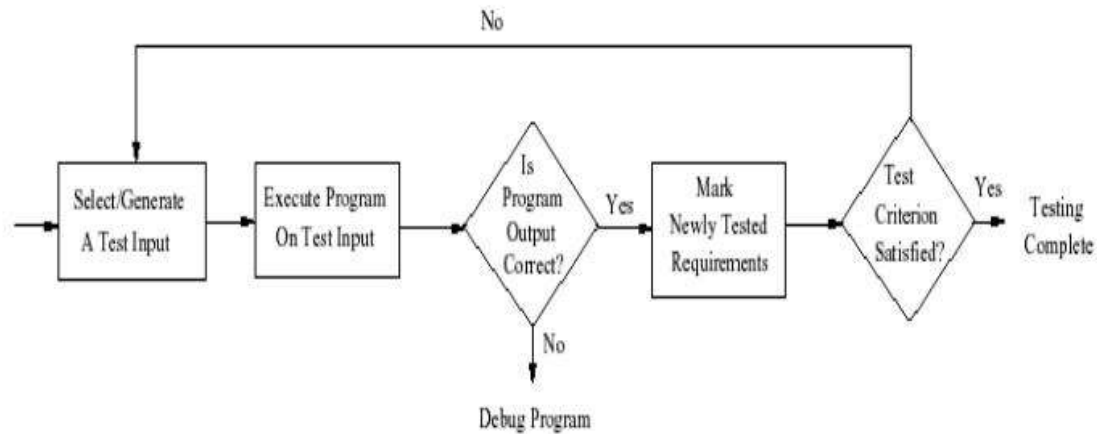
Test information flow

Three types of information flow

- Upward information flow helps the management make informed decisions,
- downward information flow ensures alignment and consistency in operations,
- lateral information flow promotes collaboration and coordination among different teams

Test information flow

The testing process of Data Flow



Test information flow

Data flow in testing

Data flow testing is a white-box testing technique that examines the data flow with respect to the variables used in the code. It examines the initialization of variables and checks their values at each instance.

Test information flow

The principle of information flow

This is much more than simple information exchange: instead of “predict and control”, the guiding principle of information flow is “sense and response”

Testing lifecycle

The Software Testing Life Cycle (STLC) refers to a set of steps or phases followed during the testing process of software development. It outlines the sequence of activities to ensure that the software meets the desired quality standards and is free from defects.

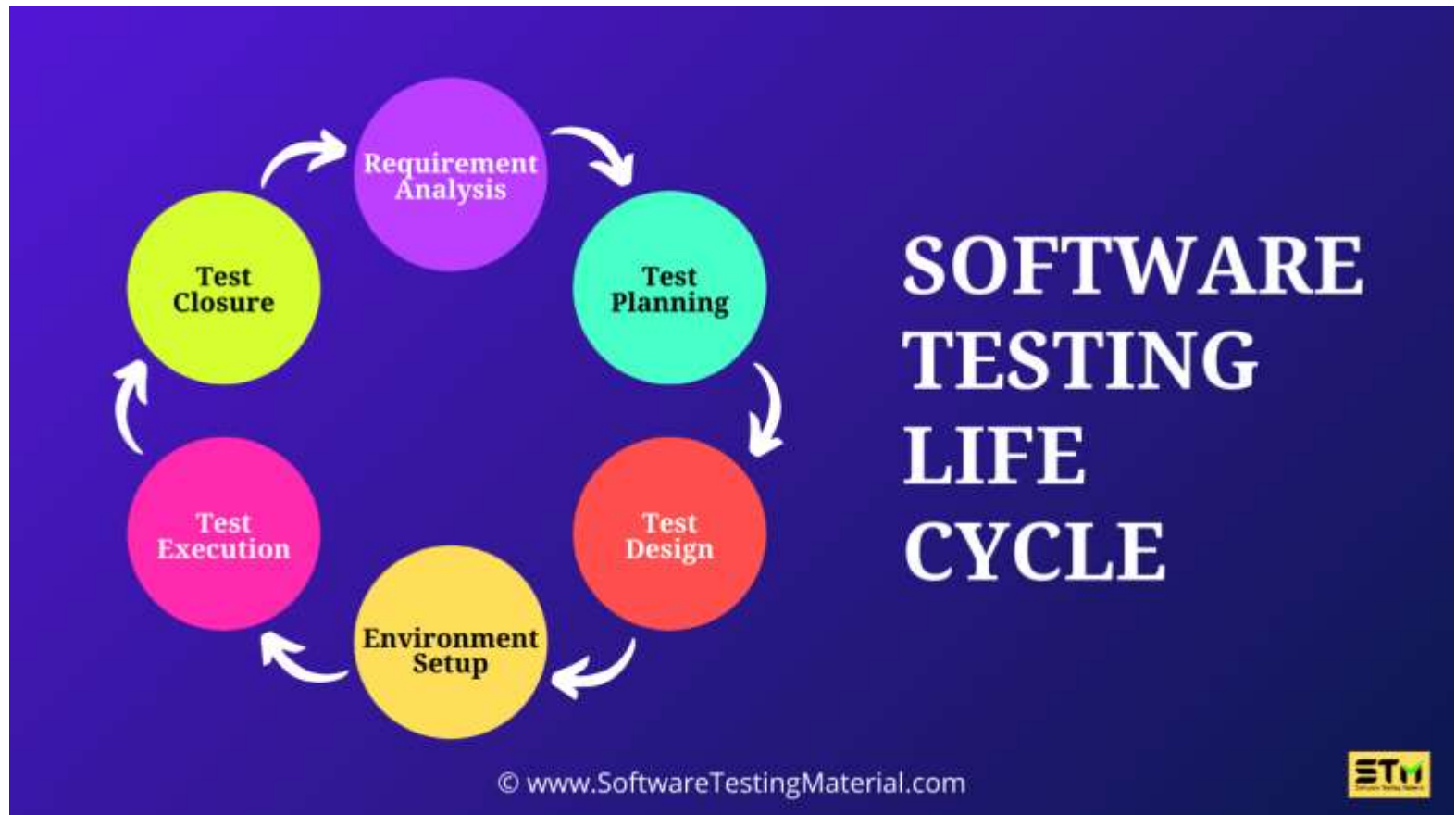
Testing lifecycle



Testing lifecycle

The 7 phases of the Software Testing Life Cycle (STLC) are: Requirement Analysis, Test Planning, Test Case Development, Test Environment Setup, Test Execution, Test Reporting, and Test Closure.

Testing lifecycle



Test Cases

What is a test case ?

A test case is a set of actions performed on a system to determine if it satisfies software requirements and functions correctly.

Test Cases

The types of test cases

8 Types of Test Cases in Software Testing

- Functionality Test Case.
- User Interface Test Case.
- Performance Test Case.
- Integration Test Case.
- Usability Test Case.
- Database Test Case.
- Security Test Case.
- User Acceptance Test Case.

Test Cases

Standard Test Case Format: **Test Case Format:**

- Test Case ID
- Test Scenario
- Test Steps
- Prerequisites
- Test Data
- Expected/Intended Results
- Actual Results
- Test Status - Pass/Fail



Test Cases

- **Test Case ID:** #BST001
- **Test Scenario:** To authenticate a successful user login on Gmail.com
- **Test Steps:**
 - The user navigates to Gmail.com.
 - The user enters a registered email address in the 'email' field.
 - The user clicks the 'Next' button.
 - The user enters the registered password.
 - The user clicks 'Sign In.'
- **Prerequisites:** A registered Gmail ID with a unique username and password.
- **Browser:** Chrome v 86. Device: Samsung Galaxy Tab S7.
- **Test Data:** Legitimate username and password.
- **Expected/Intended Results:** Once username and password are entered, the web page redirects to the user's inbox, displaying and highlighting new emails at the top.
- **Actual Results:** As Expected
- **Test Status – Pass/Fail:** Pass



Test Cases

Field	Value
Test Case ID	TC_Login_01
Test Case Name	Valid login with correct username and password
Module/Feature	Login
Pre-Conditions	User is on the login page
Test Steps	1. Enter valid username 2. Enter valid password 3. Click on the "Login" button
Test Data	Username: john_doe Password: 12345678
Expected Result	User should be redirected to the homepage/dashboard
Actual Result	(To be filled after execution)
Status	(Pass/Fail)
Remarks	(E.g., Loading time is slow)



Test Cases

How to run a test case?

To start Test Case Execution: Navigate to Project > Test Management > Test Executions. Click on the target Test Execution. Choose Test Case Execution.

Testing Team / OS-53

TP - Execution

6.7%

TE executed: 1/15 RTM Environment: Chrome TE Start Date: 2021-09-24 TE End Date: 2021-09-24 Result: **IN PROGRESS**

Test Cases (Executions) Details Relations

Assignee: All Result: all Priority: all RTM Environment: all

Order	Summary	Priority	Assignee	Result	RTM Environment	Defects
3	1 App Reminders and Newsletters	Medium	Mike Jones	Failed with test script	OS-55	
	2 Blocking app due to login 3 times failed - positive	High	Mike Jones	In all		



Designing test cases

How do you design the test case?

A test case design comprises the following elements:

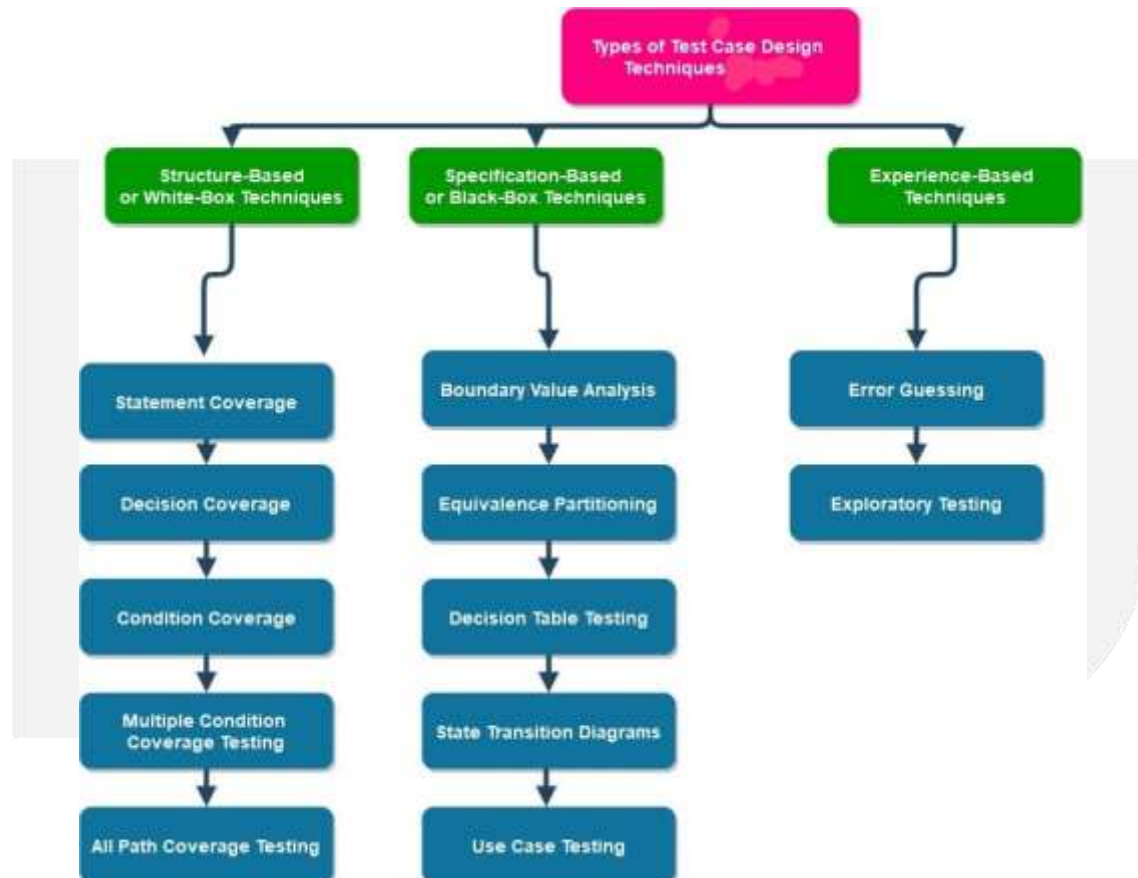
- Test objective: Every test has a purpose. ...
- Test input or precondition: Identify what data or actions are required to execute the test. ...
- Test coverage criteria: Identify which aspects of your software should be tested for completeness

Designing test cases

Why is test case design important?

Test case design techniques are essential in software testing for building robust and comprehensive test suites. These techniques help systematically explore diverse input combinations and system states, enabling efficient identification of potential defects and ensuring software quality.

Designing test cases





QUESTIONS

Basic Interview Questions

- What is software testing? Why is it important?
- What are the primary objectives of software testing?
- Differentiate between error, defect, and failure.
- Explain the difference between validation and verification.
- What is the Software Testing Life Cycle (STLC)? Name its main phases.
- What is a test case? What are the essential elements of a good test case?
- What do you mean by test information flow? How does it work in a software project?
- Why is test planning important in the software testing lifecycle?
- What is the difference between static and dynamic testing?
- What are entry and exit criteria in STLC? Provide examples.

QUESTIONS

What is software testing?

List the primary objectives of software testing.

Why is testing an important phase in the software development lifecycle?

Explain the concept of "verification" and "validation" in testing.

What is the role of a test plan in the software testing process?

How does testing contribute to software quality assurance (SQA)?

What are the main objectives of regression testing?

Compare functional testing and non-functional testing.

Evaluate the effectiveness of automated testing tools in large-scale projects.

What are the challenges in defining testing objectives in agile development?



QUESTIONS

What is the purpose of testing in software development?

List the types of software testing.

Define a testing objective.

Explain the difference between functional and non-functional testing.

What role does testing play in software quality assurance (QA)?

Why is it important to define testing objectives early in the software development process?

How would you apply testing objectives to create a test plan for a web application?

Apply the concept of regression testing to a scenario where a new feature is added to an application.

How can you use testing objectives to prioritize test cases?

Analyze the impact of not having well-defined testing objectives in a large-scale project.

How would you break down the process of creating a testing objective for a new software feature?

Compare the testing objectives of system testing and acceptance testing.

Evaluate the importance of automating test cases based on testing objectives.

Critique the effectiveness of only using manual testing to achieve testing objectives in a modern software environment.

Design a set of testing objectives for a healthcare application that manages patient records.

Propose a testing strategy that aligns with the objectives of an agile development project.

Create a set of test objectives for a mobile application that needs to be compatible with multiple devices and operating systems.

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